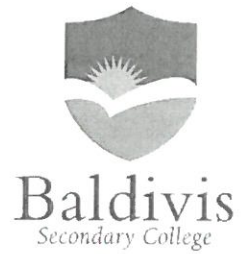


Year 8



End of Year Test

Name: Solutions

Total Marks: 45 marks
Total Time: 50 minutes

Show working and answers on this sheet. Calculator and 1 page of A4 notes.

1. (a)

[1/2 each = 5]

FRACTION	DECIMAL	PERCENTAGE
$\frac{1}{2}$	0.5 ✓	50% ✓
$\frac{1}{5}$ ✓	0.2	20% ✓
$\frac{1}{4}$ ✓	0.25 ✓	25%
$\frac{1}{1}$ ✓ * Any reasonable answer.	1.00	100% ✓
$\frac{3}{5}$	0.6 ✓	60% ✓

$(-\frac{1}{2})$ total if not simplified.
[1, 1, 2 = 4]

2. Simplify the ratios

(a) 10 : 5

(b) 7 : 21

(c) 1m : 25cm

2 : 1 ✓

1 : 3 ✓

100 : 25 ✓
⇒ 4 : 1 ✓

3. Share \$230 in a ratio of 1 : 3 : 6 to determine what the largest share will be.

[3]

$$\frac{230}{10} = 23 \quad \checkmark$$

$$\Rightarrow 23 : 69 : 138 \quad \checkmark$$

\$138 will be the largest share. $\checkmark$

4. Solve the following equations

[1, 1, 2, 2, 2 = 8]

(a) $4 + x = 9$

$$x = 5 \quad \checkmark$$

(b) $3m = 15$

$$m = 5 \quad \checkmark$$

(c) $6t + 3 - 2t = 24$

$$\checkmark \quad 4t + 3 = 24$$

$$4t = 21$$

$$\checkmark \quad t = 5.25$$

(d) $4(3x - 2) = 112$

$$\checkmark \quad 12x - 8 = 112$$

$$12x = 120$$

$$\checkmark \quad x = 10$$

(e) $5z - 4 = 8 - z$

$$\checkmark \quad 6z - 4 = 8$$

$$6z = 12$$

$$\checkmark \quad z = 2$$

5. Graph the line $y = 2x - 1$

[2]



6. For Jane's 18th birthday party her parents are hiring a band.

[2, 2 = 4]

The two bands to choose from are:

Drummin'	\$120 per hour
Air Guitar	\$100 booking fee plus \$80 per hour

Write an expression for the total cost depending on the number of hours that they play for

a) Drummin'

$120n$



b) Air Guitar

$80n + 100$



7. The following heights of have been recorded by a group of students for some seedlings they are growing in their science experiment [1, 1, 1, 2 = 5]

5cm, 4cm, 6cm, 5cm, 3m, 4.5cm

(a) Remembering that 1m = 100cm what is the value of the outlier: 300 cm ✓

(b) Determine the mean with all entries included.

$$\frac{324.5}{6} = 54.08\text{cm} \quad (2dp) \quad \checkmark$$

(c) Determine the median with all entries included.

4 4.5 5 5 6 300 ✓
median = 5.

(d) Given that the data set has an outlier which measure of central tendance should we use and why?

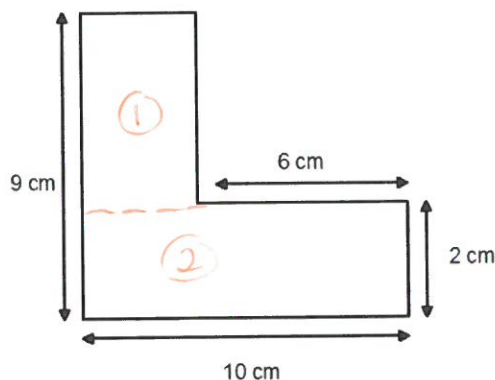
median ✓

not affected by the outlier. ✓

* Any reasonable answer

8.

[2, 3 = 5]



(a) Find the perimeter of the shape.

38cm ✓✓

(b) Find the area of the shape in mm²

① $7 \times 4 = 28\text{cm}^2$ ✓

② $10 \times 2 = 20\text{cm}^2$ ✓

Total Area = 48cm^2 ✓

$a \times 4 = 36\text{cm}^2$ ✓

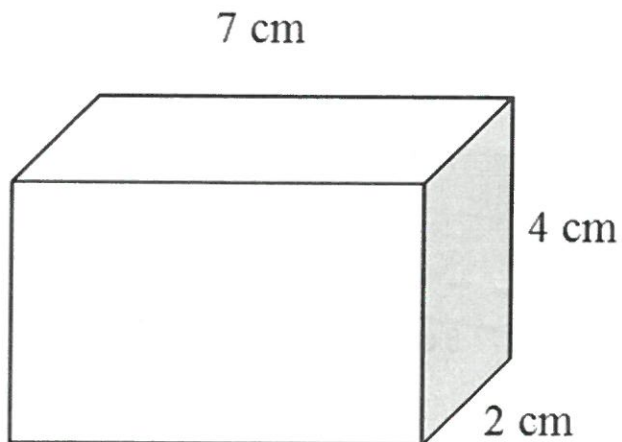
$b \times 2 = 12\text{cm}^2$ ✓

Total Area = 48cm^2 ✓

* - ! incorrect units / missing units.

9. Determine the volume of the rectangular prism below.

[2]



$$7 \times 4 \times 2 = 56 \text{ cm}^3$$

⊖! incorrect units

* unless already taken.

10. If the area of a square is 16 cm^2 , what is the side length of the square?

[1]

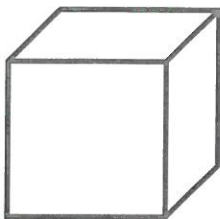


$$\sqrt{16} = 4$$

$$\therefore 4 \text{ cm}$$

11. If a cube has a volume of 8 cm^3 , what is the side length?

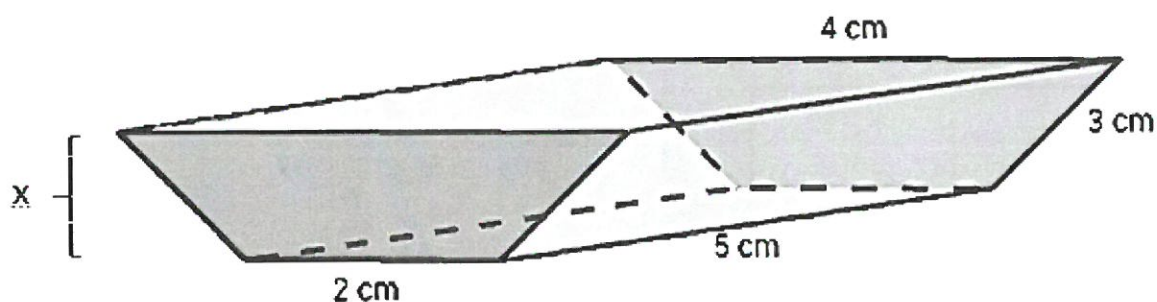
[1]



$$\sqrt[3]{8} = 2$$

$$\therefore 2 \text{ cm}$$

12. The diagram shows a prism whose base is a trapezoid. The surface area of the trapezoidal prism is 72 cm^2 . [3, 1, 1 = 5 marks]



- (a) Find the value of x .

$$72 - (2 \times 5 \times 3) - (5 \times 2) - (5 \times 4) = 12$$

$$12 \div 2 = 6 \quad \checkmark$$

$$\frac{1}{2}(2+4) \times x = 6 \quad \checkmark$$

$$3x = 6$$

$$x = 2 \text{ cm} \quad \checkmark$$

- (b) Find the volume of the prism.

$$6 \times 5 = 30 \text{ cm}^3 \quad \checkmark$$

- (c) Determine the capacity of the prism.

$$1 \text{ cm}^3 = 1 \text{ mL}$$

$$\therefore 30 \text{ mL} \quad \checkmark$$

END OF TEST